Supplementary Material for "Oxidation-Budgeted Design of Lead-Free RbGeI3 Solar Cells: Back-Surface-Field Enhancement, Realistic Loss Budgeting and SHAP Sensitivity From SCAPS-1D Optimization to Manufacturing Tolerances, Benchmarked Against CsGeI3"

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S1. One-at-a-time sweep details

Full per-parameter traces for the eleven sequential optimization steps summarized in Section V.B of the main text.

Effect of absorber layer thickness

Thickness sets the trade-off between light capture and recombination: a thicker absorber absorbs more photons and lifts J_{SC}, up to the point where the extra volume becomes a recombination liability. PCE peaks on a 0.7–0.8 μm plateau (21.21%) and eases off beyond it as recombination grows. J_{SC} climbs steadily with thickness, 25.16 to 31.29 mA/cm^2 — all below the Rühle SQ limit of 32.88 mA/cm^2 at 1.4 eV — while V_{OC} slides from 0.923 to 0.856 V. The 0.7 μm point, where the two effects balance, is the one we keep.

Effect of absorber dielectric constant

The dielectric constant shows a weak monotonic rise (23.05% at 10, 22.90 at 12, 23.19 at 15, 25.22 at 18, 25.25 at 21, 25.27 at 23.1, extending to 25.35 at 35), carried by V_{OC} at flat J_{SC}. Since ϵ_r cannot be ordered from a supplier, it is pinned at the prior mid-range value 15 with the full trace disclosed, rather than carried at a grid edge; the pinning cost is stated in the main text.

Effect of absorber electron affinity

Electron affinity lines up the energy bands between adjacent layers and so controls extraction of the charge carriers. We held everything else fixed and stepped χ from 3.7 to 4.2 eV. Peak performance lands at $\chi = 3.9$ eV with a PCE of 25.27%, bracketed on both sides (3.8: 23.57%; 4.0: 22.93%; 4.2: 18.13% with V_{OC} fallen to 0.710 V at flat J_{SC}). The 3.7 eV point fails to converge (band-offset cliff too extreme), which brackets the optimum from below by non-convergence. We stay at 3.9 eV.

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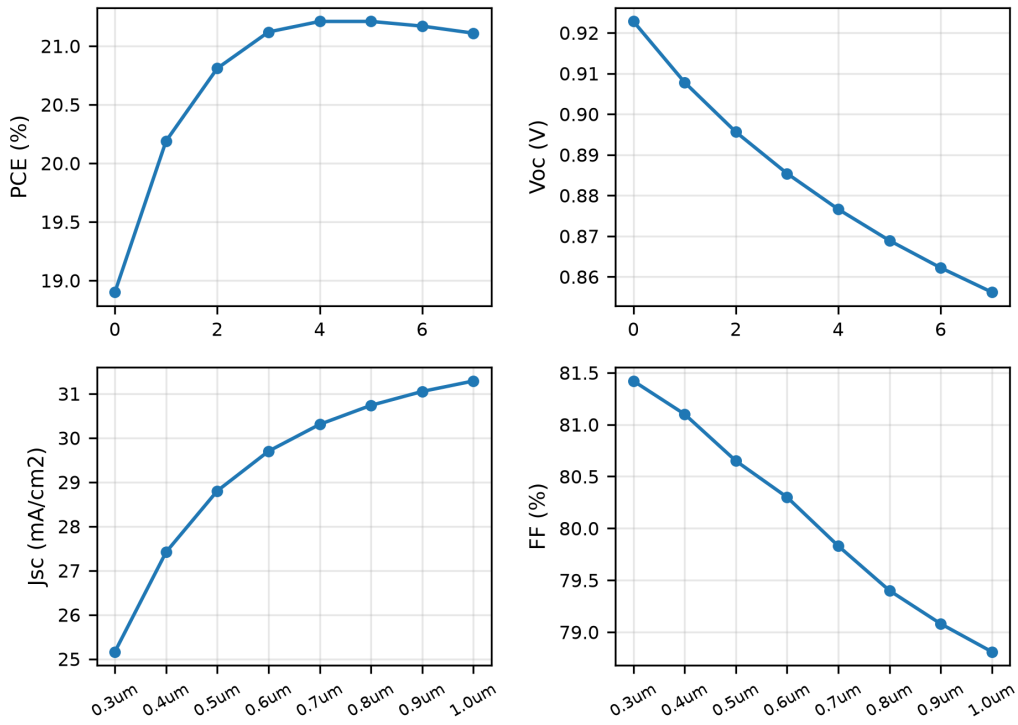


Figure S1: Variation of PCE, V_{OC}, J_{SC}, and FF with $RbGeI_3$ absorber thickness. PCE peaks at 0.7 μm, beyond which recombination losses reduce efficiency.

Alt text: Line panels of the photovoltaic parameters against absorber thickness. Efficiency peaks at 0.7 micrometres and falls beyond it as recombination losses grow.

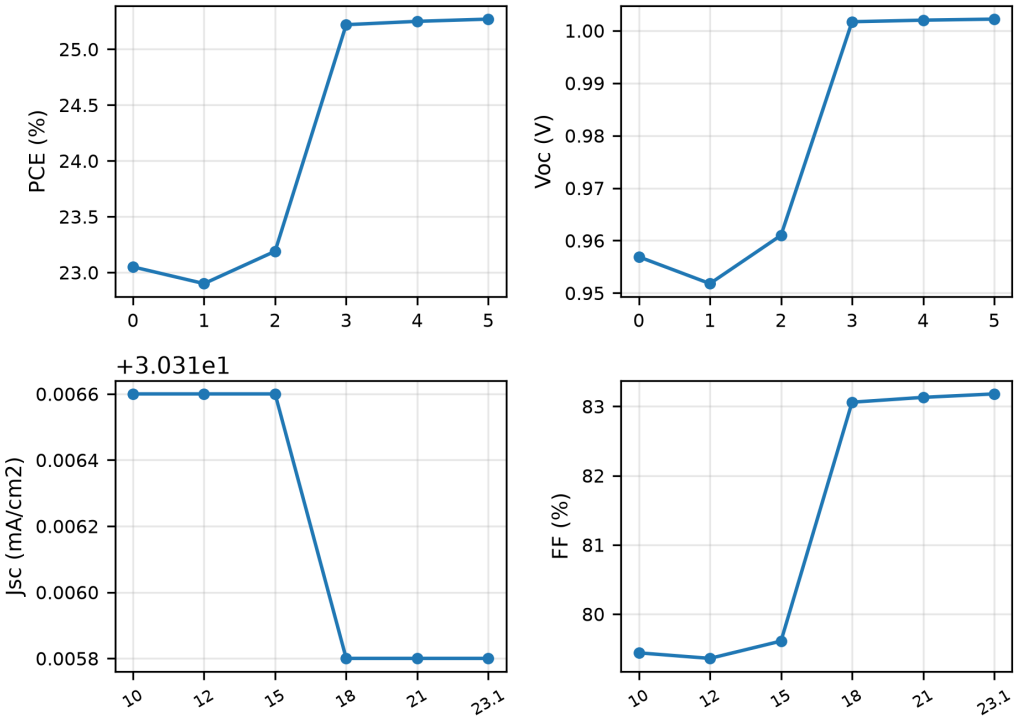


Figure S2: Variation of PCE, V_{OC}, J_{SC}, and FF with $RbGeI_3$ absorber dielectric constant. The performance is essentially insensitive to ϵr across the investigated range; $\epsilon r = 15$ is used in the final device.

Alt text: Line panels of the photovoltaic parameters against absorber thickness. Efficiency peaks at 0.7 micrometres and falls beyond it as recombination losses grow.

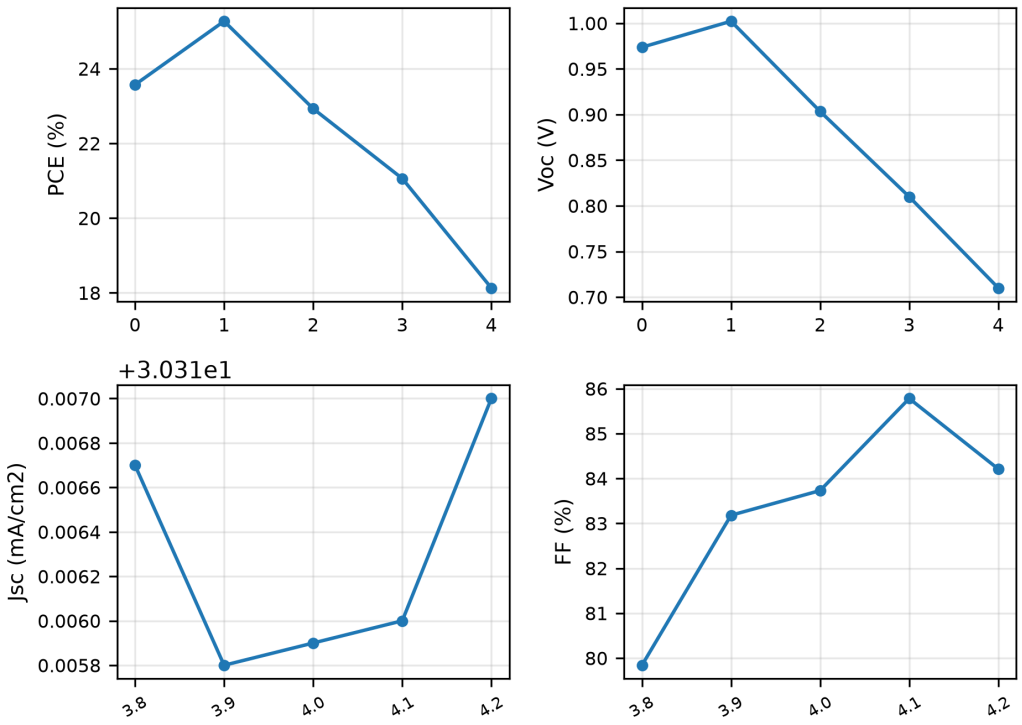


Figure S3: Variation of PCE, V_{OC}, J_{SC}, and FF with $RbGeI_3$ absorber electron affinity. PCE is highest at $\chi = 3.9$ eV due to optimal band alignment with TiO_2.

Alt text: Line panels of the photovoltaic parameters against absorber dielectric constant. All four quantities are essentially flat across the range.

Effect of TiO_2/$RbGeI_3$ interfacial defect density

The TiO_2/$RbGeI_3$ interface is where performance is won or lost. Varying its defect density from 10^{12} to 10^{18} cm^{-2} sends the device off a cliff: PCE drops from 25.27% to 10.63%, V_{OC} from 1.002 to 0.591 V and FF from 83.18 to 68.05% (Figure S4), saturating past 10^{16} cm^{-2}. More states at the junction mean more carriers captured before they reach the electrodes. Best performance comes at the unphysical 10^{12} cm^{-2} floor, but the adopted device targets 10^{14} cm^{-2}, a density a real film can hit and keep.

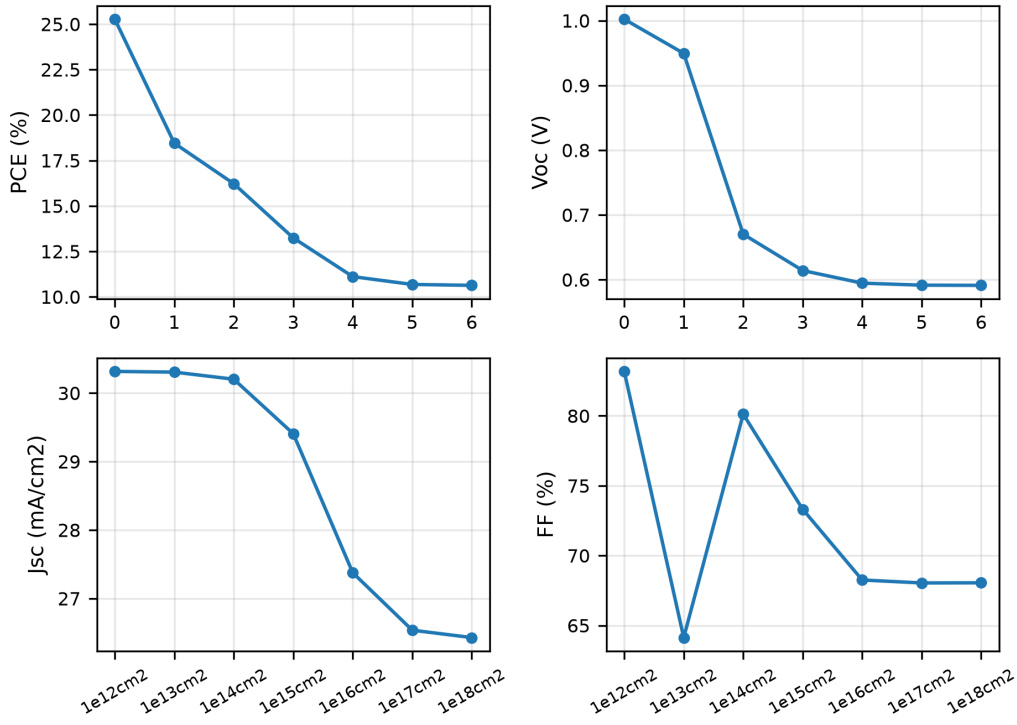


Figure S4: Variation of PCE, V_{OC}, J_{SC}, and FF with TiO_2/$RbGeI_3$ interfacial defect density. Performance degrades sharply past 10^{12} cm^{-2} and saturates past 10^{16} cm^{-2}; the $RbGeI_3$/CuI interface (Supplementary Section S1.5) is markedly more defect-tolerant.

Alt text: Line panels of the photovoltaic parameters against absorber electron affinity. Performance peaks at 3.9 electronvolts, where band alignment with the electron transport layer is optimal.

Effect of $RbGeI_3$/CuI interfacial defect density

The $RbGeI_3$/CuI junction is far more tolerant: PCE moves only from 25.27% to 19.85% across the same range (Figure S5). That contrast pins the performance limit on the TiO_2/$RbGeI_3$ interface. The adopted device still uses 10^{14} cm^{-2} here for internal consistency.

Effect of absorber bandgap

The bandgap picks which wavelengths the cell can use. Moving it from 1.3 to 1.7 eV, all else constant, raises V_{OC} from 0.966 V to 1.002 V while J_{SC} falls from 33.59 to 20.30 mA/cm^2 because a wider gap throws away more of the spectrum (1.4: 25.27%; 1.5: 22.33%; 1.6: 19.75%; 1.7: 17.52%).

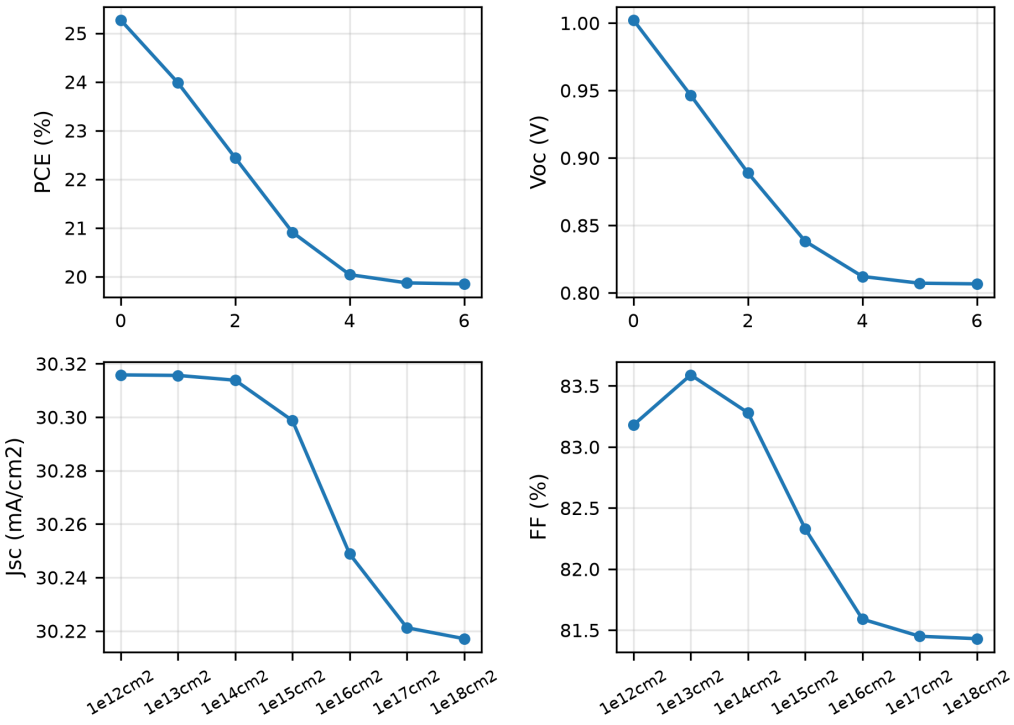


Figure S5: Variation of PCE, V_{OC}, J_{SC}, and FF with $RbGeI_3$/CuI interfacial defect density. The interface is markedly more defect-tolerant than the TiO_2/$RbGeI_3$ interface; 10^{14} cm^{-2} is used here too.

Alt text: Line panels of the photovoltaic parameters against the defect density at the titanium-oxide/rubidium interface. Performance degrades sharply beyond ten to the twelfth per square centimetre and saturates beyond ten to the sixteenth.

The raw maximum sits at 1.3 eV (26.07%), but 1.3 eV lies below the reported $RbGeI_3$ gap range and within 0.04 eV of the SQ optimum, so the SQ-margin gate rejects it and the absorber runs at 1.4 eV; the 1.2 eV point fails to converge. Rühle-tabulated SQ limits at 1.4 eV imply an ideal J_{SC} of 32.88 mA/cm^2 and a ceiling efficiency of 33.0%, beside which our 30.32 mA/cm^2 and 27.06% sit [10]. On both sides of the SQ optimum the limiting efficiency declines, steeply for wider bandgaps because fewer photons are absorbed [10, 11]: a 2.0 eV absorber would shift its absorption edge to $\lambda c \approx 620$ nm and exclude most of the visible spectrum, and our sweep already shows J_{SC} falling to 20.30 mA/cm^2 at 1.7 eV, whereas at 1.3 eV the higher J_{SC} (33.59 mA/cm^2) does not offset the V_{OC} loss (0.966 V). The corresponding cutoff wavelength is $\lambda c = 1240 / 1.4 \approx 886$ nm, consistent with the square-root-law absorption edge used in the generation profile of the main text. A targeted re-simulation varying only the absorption-grading endpoints (graded 885.7–1033.3 nm versus flat 885.7–885.7 nm, all else fixed) changes J_{SC} by 0.00 mA/cm^2, and the consolidated J_{SC} of 30.32 mA/cm^2 lies below the SQ limit of ≈33 mA/cm^2 at 1.4 eV, so the headline result is admissible with uniform square-root-law absorption at the electrical bandgap; the 27.06% device is therefore reported as an ideal-coupling upper bound, with a quantified realistic window given in Sections V.G and VII.A of the main text.

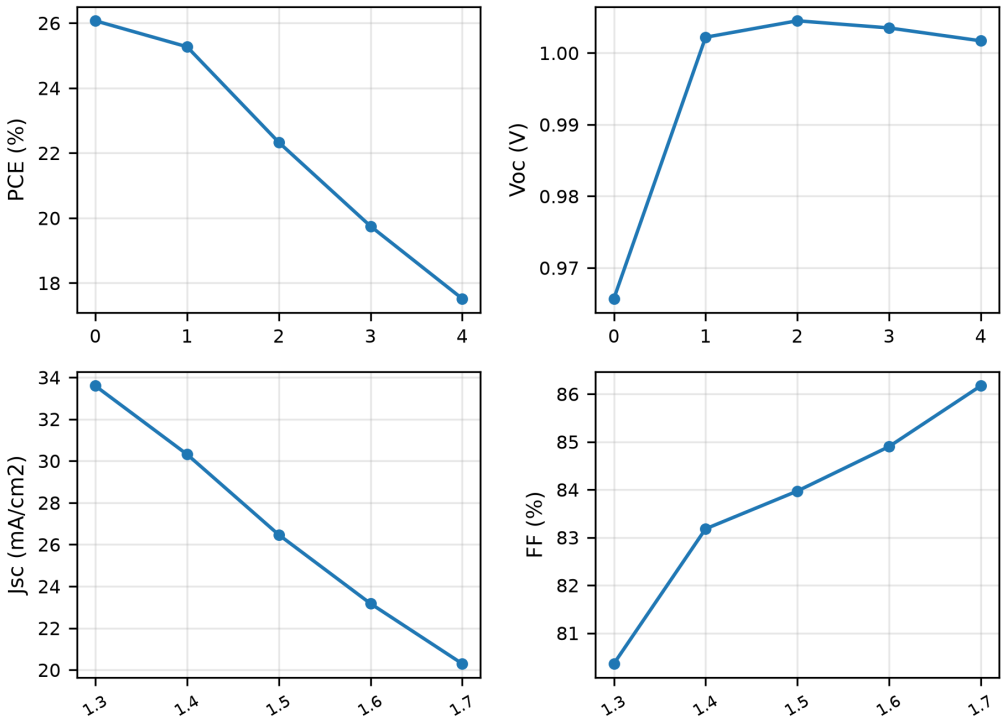


Figure S6: Variation of PCE, V_{OC}, J_{SC}, and FF with $RbGeI_3$ absorber bandgap. The raw peak at 1.3 eV is rejected by the SQ-margin gate; 1.4 eV is adopted.

Alt text: Line panels of the photovoltaic parameters against the defect density at the rubidium/copper-iodide interface. The interface is markedly more defect-tolerant; ten to the fourteenth per square centimetre is used.

Effect of TiO_2 ETL thickness

Thickness of the TiO_2 ETL cuts across electron extraction, transport, and optical transmission at once. Sweeping it from 10 to 90 nm, all else fixed, gives the best PCE at the thinnest layer: 29.09% at 10 nm, a sharp local low of 21.39% at 40 nm (FF collapses to 70.24%, reproduced on re-run — a genuine solver anomaly disclosed, not hidden), and a partial recovery to 25.33% at 90 nm (20 nm: 28.19%, cost of a pinhole-safe choice ≈0.9 pp). Planar SCAPS studies stick to thin (≤50 nm) ETLs for the same reason we do: thicker ones add series resistance and soak up photons.

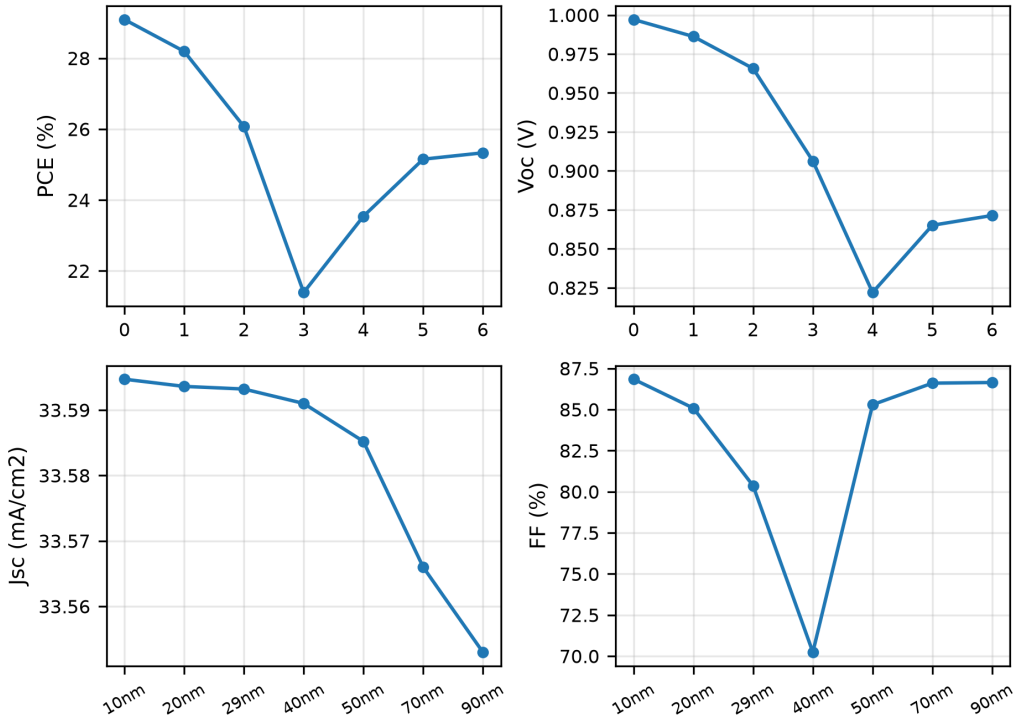


Figure S7: Variation of PCE, V_{OC}, J_{SC}, and FF with TiO_2 ETL thickness. PCE is highest at the thinnest ETL (10 nm), reflecting the shorter electron transport path.

Alt text: Line panels of the photovoltaic parameters against absorber bandgap. The raw maximum at 1.3 electronvolts is rejected by the margin gate and 1.4 is adopted.

Effect of CuI HTL thickness

CuI thickness, swept from 0.05 to 1.9 μm, leaves PCE bit-identical at 29.09% at every point; V_{OC}, J_{SC}, and FF are constant to four figures. That flatness lets us pick 100 nm purely to minimize material use. It sits at the low end of the 0.1–1.0 μm range comparable studies explored, which also reported a weak dependence.

Effect of conduction-band effective density of states N_C

Raising N_C from 1×10^{18} to 1×10^{20} cm^{-3} pushes PCE monotonically down from 30.01% to 27.60%; V_{OC} carries the loss (1.037 to 0.950 V) while J_{SC} sits bit-flat at 33.59 mA/cm^2. A lower N_C means a lower intrinsic carrier density and therefore a higher V_{OC}, so the bottom of the literature-floor

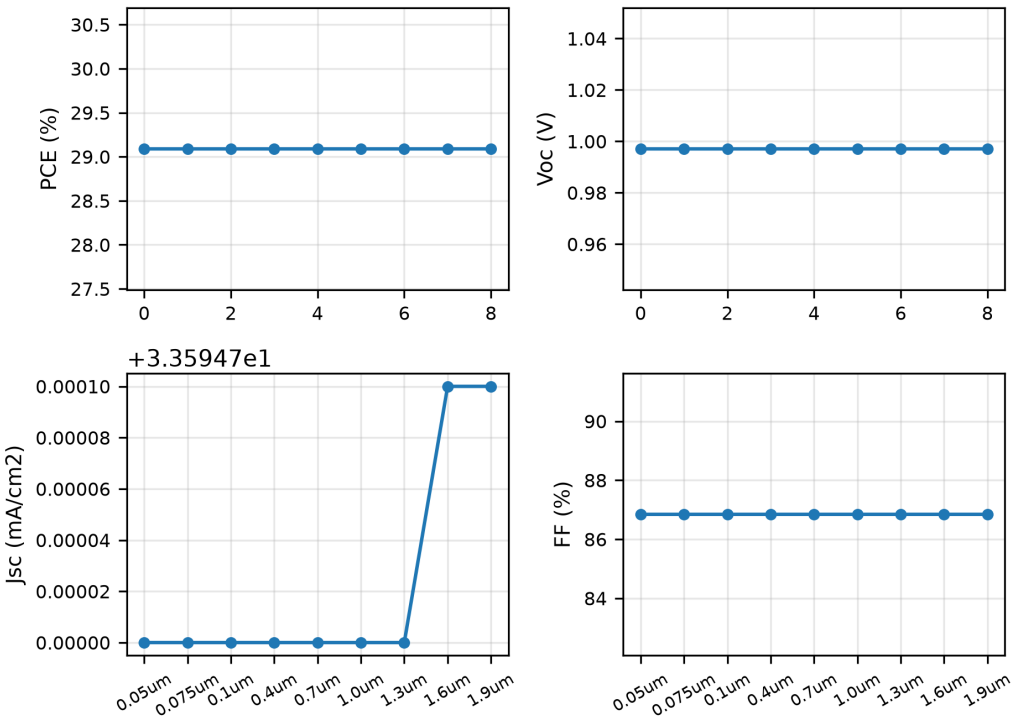


Figure S8: Variation of PCE, V_{OC}, J_{SC}, and FF with CuI HTL thickness. The performance is essentially independent of HTL thickness; 100 nm minimizes material usage.

Alt text: Line panels of the photovoltaic parameters against electron-transport-layer thickness. Performance peaks at the thinnest ten-nanometre layer, reflecting the shorter electron path.

sweep, 1×10^{18} cm^{-3}, goes into the adopted device; probing one decade below the floor gains only 0.13 pp, so the floor pin stands.

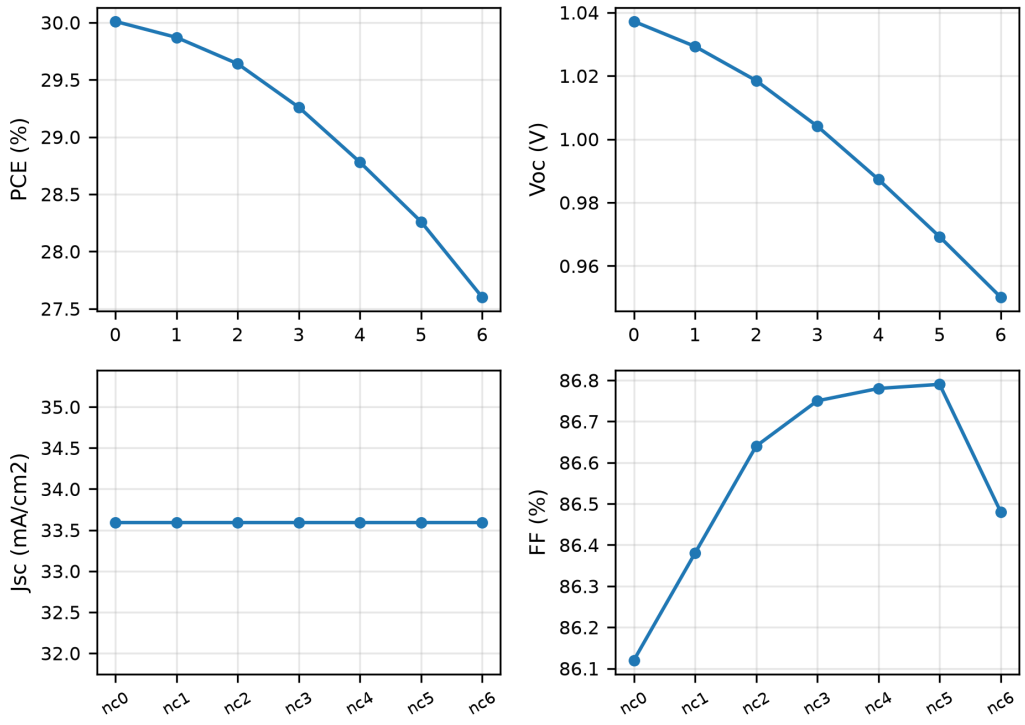


Figure S9: Variation of PCE, V_{OC}, J_{SC}, and FF with conduction-band effective density of states N_C. Lower N_C is preferred; the adopted literature-floor value is 10^{18} cm^{-3}.

Alt text: Line panels of the photovoltaic parameters against hole-transport-layer thickness. Performance is essentially flat, so 100 nanometres minimizes material use.

Effect of valence-band effective density of states N_V

N_V behaves like N_C but hits harder. Across 1×10^{18} to 1×10^{20} cm^{-3}, PCE drops from 31.57% to 29.05% as V_{OC} slides from 1.074 to 1.010 V with J_{SC} flat at ≈33.59 mA/cm^2. The smallest value wins again: $N_V = 1\times10^{18}$ cm^{-3} is carried into the adopted set; sub-floor probing gains only 0.04 pp.

D2 like-for-like pipeline (TiO_2/NiO stack)

D2 ran the identical eleven-step chain on its own NiO deck (same grids, raw-maximum carry-forward, same triple-gate assembly). Step maxima (carried): thickness 0.8 μm (21.30%); bulk N_t 10^{12} cm^{-3} (26.46%); ϵ_r 23.1 (26.46%); χ 3.9 (26.46%, bracketed by non-convergent 3.7/4.2); TiO_2-side N_t 10^{12} cm^{-2} (26.46%); NiO-side N_t 10^{12} cm^{-2} (26.46%, degrading gracefully to 22.81% at 10^{18}); E_g 1.4 eV (26.46%); ETL 10 nm (30.16%, with the 40 nm anomaly reproduced at 22.44%); HTL flat at 30.16%; N_C 10^{18} cm^{-3} (30.38%); N_V 10^{18} cm^{-3} (32.87%). Gate-adopted D2: 28.16% (1.1116 V, 30.3154 mA/cm^2, 83.57%). Archives: `r8_D2_s_*.json`, chain record `r8_D2_chain.json`, deck `r8_D2_H1_adopted.def`, J–V `r8_D2_H1_JV.json`.

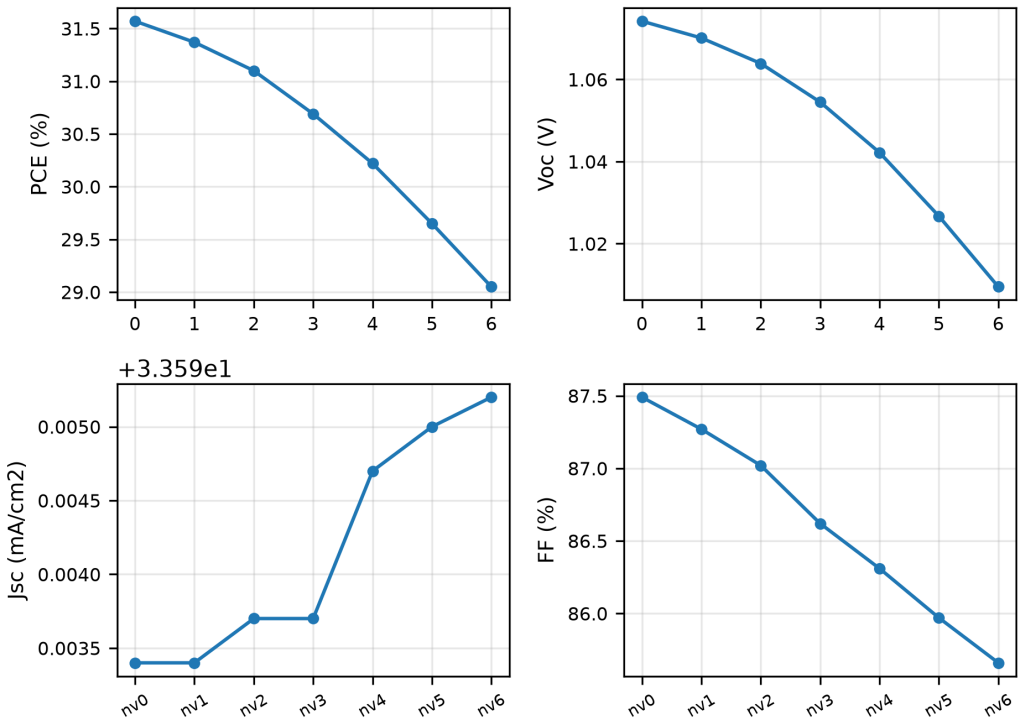


Figure S10: Variation of PCE, V_{OC}, J_{SC}, and FF with valence-band effective density of states N_V. The effect is more pronounced than for N_C; the adopted literature-floor value is 10^{18} cm^{-3}.

Alt text: Line panels of the photovoltaic parameters against conduction-band density of states. Lower values are preferred and the literature-floor value of ten to the eighteenth is adopted.

S2. Illumination and dark characteristics

Illumination dependence

The optimized cell was also driven under 0.1 to 1.5 suns (Figure S11). J_{SC} tracks intensity almost linearly, the natural consequence of carrier generation scaling with photon flux, while V_{OC} rises logarithmically. The slope of that logarithmic rise gives an ideality factor n ≈ 1.17, so deep-defect Shockley-Read-Hall recombination is not dominant under operating conditions, and the cell stays efficient across the entire range. Re-simulation on the rebased deck (0.1–1.5 suns, archived `r8_darkillum.json`) gives J_{SC} = 3.03/9.09/15.16/24.25/30.32/45.47 mA/cm^2 (perfect linearity) and n = 1.19 from the V_{OC}–ln(I) slope over 0.3–1.5 suns, confirming the reported n ≈ 1.2.

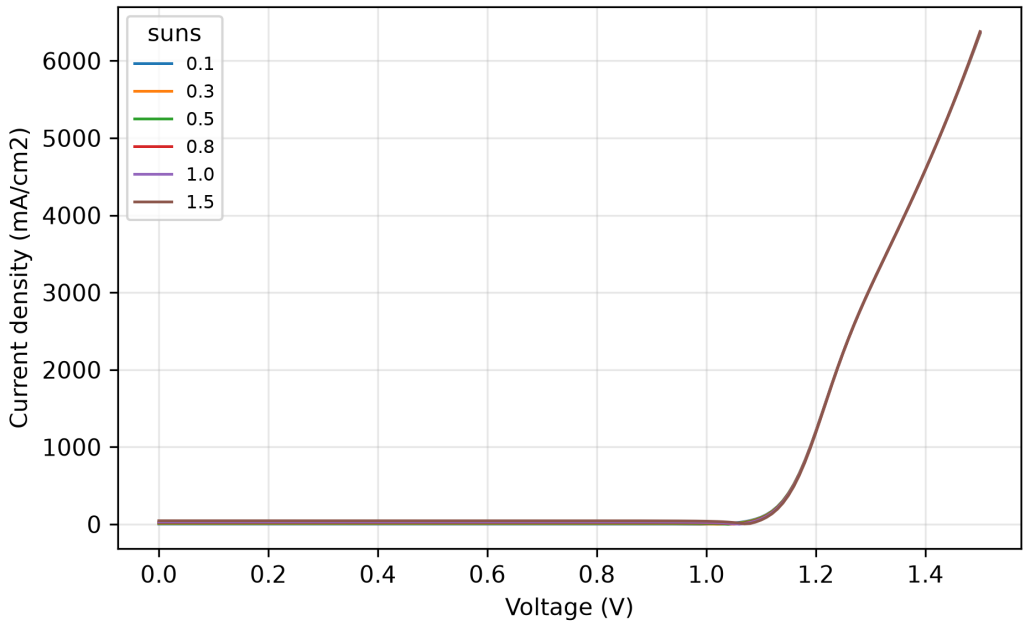


Figure S11: Variation of photovoltaic parameters of the optimized device with illumination intensity from 0.1 to 1.5 suns.

Alt text: Four-panel plot of the optimized device versus illumination intensity from 0.1 to 1.5 suns. Current rises almost linearly with light level while voltage rises logarithmically.

Dark J–V characteristics

Under dark bias (Figure S12, archived `r8_dark_JV.json`) the cell rectifies hard: forward current at +1 V exceeds reverse current at −0.5 V by a factor of 1.7×10^{10}, and leakage stays under 10^{-10} A/cm^2 across the whole reverse window. The low-bias exponential forward region (0.2–0.6 V, $r = 0.9986$) fits to an ideality factor n ≈ 1.2 and J0 ≈ 6×10^{-11} A/cm^2; higher-bias windows bend upward from series/high-injection effects and are not used. Together these numbers describe high-quality bulk and interfaces with no measurable shunts.

Study (Year) [Ref.]	Device Structure	PCE (%)	Key Limitation
Pindolia et al. (2022) [2]	FTO/TiO_2/$RbGeI_3$/NiO/Ag	9.77	Broad multi-parameter sweep (HTL, ETL, thickness, doping, defects, back contact, temperature); baseline efficiency remains modest and Ge^{2+} oxidation not addressed.
Saikia et al. (2022) [7]	FTO/TiO_2/$CsGeI_3$/CuI/Au	10.80	No multi-HTL comparative analysis.
Bouazizi et al. (2022) [3]	FTO/PCBM/ CsSn0.5Ge0.5I3/Spiro/Au	18.13	Mixed Ge/Sn absorber; thermal sensitivity.
Loumachi et al. (2024) [5]	ITO/C_{60}/$RbGeI_3$/CBTS/Ag	27.21	Optimized $RbGeI_3$ thickness and series/shunt resistance; interface defect optimization not addressed.
Raj et al. (2025) [8]	FTO/ TiO_2/$RbGeI_3$/PEDOT:PSS/ Au	18.60	Organic HTL limits long-term stability.
Pathak et al. (2026) [6]	FTO/TiO_2/$RbGeI_3$/Spiro-OMeTAD/Au	26.47	$Ge^{2+} \rightarrow Ge^{4+}$ oxidation not addressed; only single ETL/HTL combination studied.
Dash et al. (2025) [4]	DFT + SCAPS-1D of $RbGeX_3$ (X=Cl, Br, I)	25.76	Primarily material-level comparison; limited device-architecture optimization.
Samaki et al. (2023) [1]	FTO/TiO_2/$RbGeI_3$/Spiro/Ag	18.10	HTM screening that identified CuI; single-architecture optimization.

Table S1: Summary of recent SCAPS-1D studies on lead-free Ge-based and Sn-based perovskite solar cells.

Alt text: Survey table listing eight recent simulations of germanium- and tin-based lead-free solar cells with device structure, efficiency and the key limitation of each study.

Parameter	TiO_2	FTO	$RbGeI_3$	CuI
Thickness (nm)	30	500	400	100
Bandgap E_g (eV)	3.2	3.2	1.4	3.1
Electron affinity χ (eV)	4.0	4.4	3.9	2.1
Dielectric permittivity ϵr	9	9	23.01	6.5
CB effective DOS N_C (cm^{-3})	2×10^{18}	2.2×10^{18}	1×10^{17}	2.8×10^{19}
VB effective DOS N_V (cm^{-3})	1.8×10^{19}	1.8×10^{19}	1×10^{17}	1×10^{19}
Electron mobility μe (cm^2 V^{-1} s^{-1})	20	20	28.6	100
Hole mobility μp (cm^2 V^{-1} s^{-1})	10	10	27.3	43.9
Donor density N_D (cm^{-3})	9×10^{16}	1×10^{19}	1×10^{9}	0
Acceptor density N_A (cm^{-3})	0	0	1×10^{9}	1×10^{18}
Bulk defect density N_t (cm^{-3})	1×10^{15}	1×10^{14}	1×10^{15}	1×10^{15}

Table S2: Initial electrical and optical parameters used in the SCAPS-1D simulation of the proposed FTO/TiO_2/$RbGeI_3$/CuI/Au solar cell.

Alt text: Parameter table giving thickness, bandgap, affinity, permittivity, densities of states, mobilities, doping and defect densities for the titanium oxide, fluorine-doped tin oxide, rubidium and copper-iodide layers.

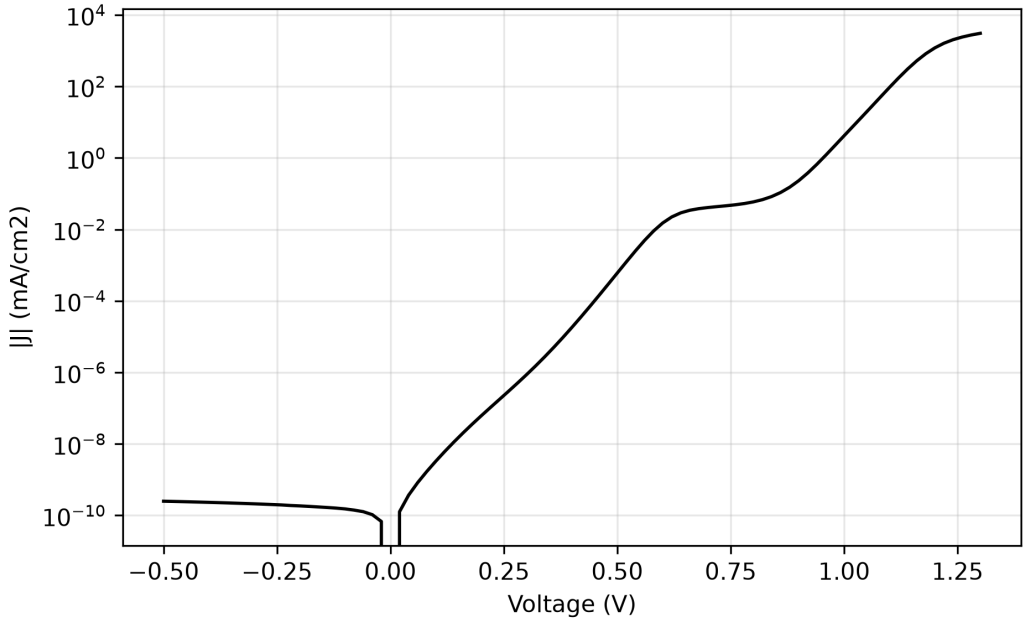


Figure S12: Dark J–V characteristic of the optimized device. The forward branch is exponential over nearly three decades; the reverse branch remains below 10^{-10} A/cm^2.

Alt text: Semilogarithmic dark current-voltage curve with an exponential forward branch spanning nearly three decades and reverse leakage below ten to the minus-ten amperes per square centimetre.

Interface	Interface defect density N_t (cm^{-2})
TiO_2 / $RbGeI_3$	1×10^{14}
$RbGeI_3$ / CuI	1×10^{14}

Table S3: Initial interface defect density applied at each heterojunction.

Alt text: Short table listing one ten-to-the-fourteenth per-square-centimetre defect densities applied at each of the two heterojunctions.

Parameter	Value
Simulation software	SCAPS-1D version 3.3.10
Illumination spectrum	AM 1.5G
Incident power density	100 mW/cm^2 (1000 W/m^2)
Operating temperature	300 K (27 °C)
Illumination intensity	1 Sun
Scan mode	Illuminated J–V
Device type	Planar heterojunction
Front contact	FTO
Back contact	Au
Initial material parameters	Table S2
Initial interface defect density	Table S3

Table S4: Simulation conditions used in SCAPS-1D.

Alt text: Key-value table of the simulation settings, including the solver version, the standard sunlight spectrum, one-sun intensity, room temperature and the front and back contact metals.

BSF d (nm) × N_A (cm^{-3})	10^{17}	10^{18}	10^{19}
30	27.55	27.93	28.00
50	27.59	27.93	28.00
80	27.60	27.94	28.01
100	27.60	27.94	28.01

Table S5: BSF grid PCE (%; `r8_bsf.json`, 12/12 converged) on the rebased H1-adopted deck. Doping dominates; thickness flat within 0.01 pp at N_A 10^{19} cm^{-3}.

Alt text: Four-by-three grid of efficiency values over back-surface-field thickness and doping. Values rise with doping from about 27.6 to 28.0 percent while thickness has almost no effect.

Waterfall node	V_{OC} (V)	J_{SC} (mA/cm^2)	$PCE_{1\text{-sun}}$ (%)
H0 adopted (H1)	1.0613	30.32	27.06
W1 +10% reflection (90 mW/cm^2)	1.0600	27.28	24.28
W2 +bulk 10^{15} cm^{-3} (90 mW/cm^2)	1.0200	27.28	21.93
S3 +interfaces 10^{15} cm^{-2} (90 mW/cm^2)	0.9800	27.27	20.78
S4 BSF on S3 stack (90 mW/cm^2)	1.0100	27.27	21.76

Table S6: Stacked loss waterfall (`r8_waterfall_JV.json` for W1/W2, `r8_S3_JV.json`, `r8_s490_JV.json`). Sub-1-sun points report $PCE_{1\text{-sun}} = VJFF/100$ explicitly since SCAPS normalizes `eta` to the set illumination.

Alt text: Five-row table stepping the efficiency from the 27.06 percent ideal device through reflection, bulk and interface losses to 20.78 percent, recovering to 21.76 percent with the back-surface field.

S3. Screening and simulation setup tables

S4. BSF, junction baseline and ML-surrogate notes

BSF structure

The BSF variant inserts a Cu_2O-like p^+ layer (E_g 2.1 eV, χ 3.2 eV, ϵ_r 7.5, N_A 10^{17}–10^{19} cm^{-3}, 30–100 nm) between $RbGeI_3$ and CuI, splitting that interface into CuI/BSF and BSF/$RbGeI_3$, both carrying the same neutral-trap model as the interface they replace. BSF defs branch from the rebased H1-adopted deck, so the radiative correction is already in the base. Generating script: `r8_bsf.py`; grid in `r8_bsf.json`; ideal-BSF J–V in `r8_BSF_JV.json`; BSF-on-realistic-stack J–V in `r8_s490_JV.json`. Trap attribution runs (`r8_DX_*.def`, `r8_DX_a/b/c.json`): rear-10^{16} on H1 gives 19.65%; on BSF50-NA19 with both rear blocks raised gives 24.36%, with only the absorber-side block raised also 24.36%, and with only the far CuI/BSF block raised 28.00% (no penalty). The p^+ no-offset control (BSF layer reset to CuI bands, N_A kept) gives 27.82% ideal and 21.33% trapped.

Junction baseline

Closed-form parameters anchored to the rebased H1 working point (scripted C–V capture returns header-only under v3.3.10 scripting, so no SCAPS C–V is claimed; V_{bi}/depletion-width/Mott–Schottky claims are withdrawn since the near-intrinsic absorber is fully depleted): n_i 1.75×10^{12} m^{-3} (from N_C=N_V 10^{18} cm^{-3}, E_g 1.4 eV); SRH τ 1000 ns at 10^{14} cm^{-3}; L 8.6 μm $\gg$ 700 nm absorber; α 3.3×10^6 m^{-1}.

ML surrogate

XGBoost/Random-Forest surrogates on the 200-point LHS-UQ block (8 features: $E_g, N_t, N_C, N_V, \mu_n, \mu_p, N_{\text{int1}}, N_{\text{int2}}$ ±25% around the optimum, 80/20 split, seed 7): XGB R^2 0.9074, RMSE 0.57; RF R^2 0.9067. SHAP ranking: bulk $N_t > E_g > \mu_n >$ CuI-side $N_{\text{int2}} > TiO_2$-side $N_{\text{int1}} > N_V > \mu_p > N_C$. Generating script: `r8_ml.py`; metrics in `r8_ml.json`; UQ block in `r8_uq.json`.

S5. Experimental anchoring: methods and provenance

Reference device

Wang et al., Science 365, 591–595 (2019): FTO/c-TiO_2/β-$CsPbI_3$ (350–400 nm)/spiro-OMeTAD/Ag, AM1.5G, J–V at ∼45 °C. Control 15.1% (1.05 V, 20.03 mA/cm^2, 0.72); CHI-passivated champion 18.4% (1.11 V, ∼20.0 mA/cm^2, 0.82); EQE-integrated J_{SC} 19.71/19.85 mA/cm^2; 32-device statistics.

Digitization

Fig. 4A rendered at 400 dpi; tick-regression calibration (±0.002 V, ±0.22 mA/cm^2); red/black color segmentation with blue-annotation and legend masking; column-median traces; legend-remnant filter. Self-check: extracted MPP metrics reproduce tabulated values within ∼1% (control plateau reads ∼2% low — representative curve vs champion statistics, not fitted). CSVs: `validation/wang2019_ctrl_JV.csv`, `validation/wang2019_CHI_JV.csv`.

SCAPS rebuild

Base: repo `perovskite-rbgei3.def`. Absorber → $CsPbI_3$ (E_g 1.68 eV from UV-vis edge, χ 3.8, ϵ_r 10, N_C=N_V 10^{19} cm^{-3}, μ 25/25 cm^2/Vs, 375 nm); HTL → doped spiro-OMeTAD (χ 2.3, μ 2×10^{-3} cm^2/Vs for the Li-TFSI recipe, 180 nm); c-TiO_2 40 nm; Ag back contact at effective 5.2 eV (pinned metal/organic interface; nominal 4.7 eV imposes a 0.5 eV barrier contradicted by the measured FF; 5.0/5.1 eV systematically fail to converge); sqrt-law $A = 1.5 \times 10^5$ (Jsc-calibrated once); 318 K working point. Fitted: bulk N_t × interface N only. Champion: 3×10^{13} cm^{-3} / 3×10^{13} cm^{-2}. Control: 3×10^{13} cm^{-3} / 10^{14} cm^{-2} + external $V - JR_s$ correction $R_s = 6.9\ \Omega{\cdot}\text{cm}^2$ (defects alone floor FF at ∼0.84; cracks/pinholes/hysteresis documented). Builder: `validation/r7_validate.py`; grid: `validation/r7_grid_final.json`; best-fit J–V: `validation/r7_final_JV.json`; post-R_s control J–V: `validation/r7_control_Rs69_JV.csv`; audit trail: `validation/VALIDATION_LOG.md` (includes non-converged runs).

Database benchmark

Perovskite Database Project via NOMAD REST API, Oct. 2026: 43,106 `PerovskiteSolarCell` entries; 41,388 at 1-sun (600–1200 W/m^2); strict-inorganic filter (no MA/FA/BA/PEA/AVA/GU/NH_4/sulfonium/anilinium/thiocyanate tokens) → 2,201; archive-level J–V for all 2,201 (421 $CsPbI_3$). Analysis-clean counts used in the main text (41,316 one-sun, 2,141 inorganic) drop entries with missing or out-of-range metrics ($0 < \text{PCE} < 35$, $0 < V_{OC} < 2.5$, $0 < \text{FF} \leq 1$). Fetchers: `validation/nomad_fetch.py`, `validation/nomad_archives.py`; frozen cache: `validation/nomad_jv_clean.json`. Inorganic PCE: median 7.22%, p90 13.74%, max 18.22%; $CsPbI_3$ max 17.82%; 5 Ge-bearing entries (best 10.8%), zero $RbGeI_3$.

S6. Oxidation-budget derivation and bounds

Equilibrium V_{Ge} concentration $N = N_{sites} \exp(-E^f/kT)$ with $N_{sites} = 1/a^3 = 4.87 \times 10^{21}$ cm^{-3} ($a = 5.9$ Å) at 300 K gives: $E^f = 0.46 \rightarrow 1.0 \times 10^{14}$; $0.40 \rightarrow 9.3 \times 10^{14}$; $0.35 \rightarrow 6.4 \times 10^{15}$; $0.30 \rightarrow 4.4 \times 10^{16}$; $0.15 \rightarrow 1.5 \times 10^{19}$ cm^{-3}. These land on the consolidated N_t ladder (27.06/24.40/20.76% at $10^{14}/10^{15}/10^{16}$ on the H1 deck) as plotted in main-text Fig. 24 (generator: `validation/r8_oxidation_fig.py`; case file: `validation/oxidation_case.md`). The 300 K values are lower bounds: freeze-in from 100 °C (373 K) requires $E^f \approx 0.57/0.50/0.42$ eV for $10^{14}/10^{15}/10^{16}$ cm^{-3}, and from 150 °C (423 K) 0.65/0.56/0.48 eV; processing bands in the main text are illustrative Ef assumptions. A current-state rung ($N_t = 10^{17}$ cm^{-3} on H1, `r8_budget_check.json`) gives 15.53%. Tying doping to defects ($N_t = 10^{15}$, $N_A = 2 \times 10^{15}$ cm^{-3}) gives 25.52% vs 24.40% untied. Each V_{Ge} donates two holes, so $p \approx 2[V_{Ge}]$ reproduces the N_A-collapse onset at 10^{16} cm^{-3}. SRH lifetimes $\tau^{-1} = \sigma v_{th} N_t$ give 1000/100/10/1/0.1 ns at $N_t = 10^{14}$–10^{18} cm^{-3}; reported sub-ns TRPL in Ge-containing films [12] therefore implies $N_t \gtrsim 10^{17}$, 100–1000× above the $\leq 10^{15}$ tolerance. Bounds: energetics are $CsGeI_3$ DFT transferred to $RbGeI_3$ (same Ge–I framework; A-site band-edge-inert); equilibrium is a bound on kinetic oxidation; the V_I deep-trap picture is functional-dependent, so no claim rests on V_I.

S7. Reproducibility appendix

All SCAPS results were produced with SCAPS-1D v3.3.10 scripted through `scaps-runner` (`r5_lib.py`: `BASE` deck, `run_set`, `edit_block`, `op_summary_only`). Base deck: repo `perovskite-rbgei3.def`; adopted deck: buildable as `r8_H1_adopted.def` (frozen in `REVIEW_FIX_LEDGER.md`). Run scripts and archives: OAT `r8_oat.py` → `r8_s*.json` + `r8_oat_summary.json`; extras/costs `r8_extras.py` → `r8_extras.json`; GA `r8_ga.py` → `r8_ga_results.json` (best-found 28.30% as optimizer reach below the adoption floors); BSF `r8_bsf.py` → `r8_bsf.json` (+ϵ_r extension `r8_eps2.json`); trap-attribution and p^+-control runs → `r8_DX_a/b/c/d/e.json` (rear-only 20.29/19.65%, front-only 13.12/10.77%, both 10.59% at $10^{15}/10^{16}$; far-side-only 28.00%; p^+ control 27.82% ideal, 21.33% trapped; superseding the unattributed `r8_nifsens.json` pair); D2 pipeline `r8_D2.py` → `r8_D2_s_*.json` + `r8_D2_chain.json` + `r8_D2_H1_adopted.def` + `r8_D2_H1_JV.json` (literature-standard NiO documented above, nothing fitted); stacked window `r8_stack.py` → `r8_stack.json` + `r8_S3_JV.json` + `r8_s490_JV.json`; optics waterfall nodes `r8_waterfall_JV.json` (W1 24.28%, W2 21.93%); temperature/Et/sigma/Cs `r8_supplement.py` → `r8_temp.json`, `r8_etsigma.json`, `r8_sigma.json`, `r8_cs.json`; oxidation-budget checks `r8_budget_check.json` (current-state Nt17 rung 15.53%, NA-tied Nt15 point 25.52%); UQ/ML `r8_uq.py`, `r8_ml.py` → `r8_uq.json`, `r8_ml.json`; illumination/dark `r8_darkillum.json`, `r8_dark_JV.json`. Known tooling limits, all disclosed in the main text: scripted `save results.qe` and `save results.cv` return header-only files under v3.3.10 scripting (no same-deck QE curve; no SCAPS C–V; junction claims withdrawn to lifetime-only); `save results.genrec` yields trap-occupation tables only. Non-convergent grid points (`r8_s04_chi_3.7`, `r8x_eg12`, D2 `chi_3.7/4.2`) are reported as non-convergence, not omitted silently.textttr8_ bsf.json (+ϵ_r extension `r8_eps2.json`); trap-attribution and p^+-control runs → `r8_DX_a/b/c.json` (superseding the unattributed `r8_nifsens.json` desensitization pair); stacked window `r8_stack.py` → `r8_stack.json` + `r8_S3_JV.json` + `r8_s490_JV.json`; optics waterfall nodes `r8_waterfall_JV.json` (W1 24.28%, W2 21.93%); temperature/Et/sigma/Cs `r8_supplement.py` → `r8_temp.json`, `r8_etsigma.json`, `r8_sigma.json`, `r8_cs.json`; UQ/ML `r8_uq.py`, `r8_ml.py` → `r8_uq.json`, `r8_ml.json`; illumination/dark `r8_darkillum.json`, `r8_dark_JV.json`; oxidation-budget checks `r8_budget_check.json` (current-state Nt17 rung, NA-tied Nt15 point). Known tooling limits, all disclosed in the main

text: scripted `save results.qe` and `save results.cv` return header-only files under v3.3.10 scripting (no same-deck QE curve; no SCAPS C–V; junction claims withdrawn to lifetime-only); `save results.genrec` yields trap-occupation tables only; no GUI-automation tooling (`xdotool`/`wmctrl`/`pyautogui` absent) for scripted C–V. Non-convergent grid points (`r8_s04_chi_3.7`, `r8x_eg12`, D2 `chi_3.7/4.2`) are reported as non-convergence, not omitted silently. The D2 NiO deck (`r8_D2_H1_adopted.def`, script `r8_D2.py`) substitutes literature-standard NiO (E_g 3.6 eV, χ 1.8 eV, ϵ_r 11.7, N_C/N_V $2.8 \times 10^{19}/10^{19}$ cm^{-3}, μ 12/2.8 cm^2/Vs, N_A 10^{18} cm^{-3}) for CuI with the Au contact kept as in screening; nothing in it is fitted.

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